Boosting

Agenda

- Boosting
- Bagging Vs Boosting
- AdaBoost
- Gradient Boost
- XGBoost

Let's begin the discussion by answering a few questions

Boosting Quiz

What is a key differentiation of boosting algorithms from bagging algorithms?

- A: Boosting trains all models in parallel
- B: Boosting decreases model variance
- C: Boosting increases model interpretability
- D: Boosting trains all models sequentially to improve upon the wrong predictions of the previous weak learner

Boosting Quiz

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Bagging vs Boosting

Bagging	Boosting
Creates different subsets of data using random sampling with replacement to independently train each weak learner	Trains the weak learners sequentially on the entire dataset, reweighting misclassified data points to focus on reducing errors iteratively
All weak learners are independent and can be trained in parallel	All weak learners are dependent on each other and have to be trained sequentially
Aims to decrease variance, not bias	Aims to decrease bias, not variance
E.g, Random Forests	E.g, AdaBoost, Gradient Boost

Boosting Quiz

Statement 1: Boosting builds a series of weak learners where each model attempts to correct the errors of its predecessor, often by re-weighting misclassified samples or fitting the next model to the remaining residuals.

Statement 2: Boosting trains all weak learners independently in parallel, and each data point is given equal importance throughout the training process.

Statement 3: Boosting always reduces training time because it combines all weak learners into a single model at once.

Which of the following is correct?

A. Statement 1

B. Statement 2

C. Statement 1,2

D. Statement 3

Boosting Quiz

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Boosting

Boosting is an ensemble learning technique that combines multiple weak learners to form a strong model.

It trains models sequentially, where each new learner focuses more on the mistakes made by the previous ones.

Misclassified data points are assigned higher weights, while correctly classified points receive lower weights. This process continues until a stopping criterion such as maximum iterations or desired performance is reached.

The final prediction is made using weighted averaging or weighted voting of all learners

Boosting Quiz

In AdaBoost, how does the algorithm adjust the weights of samples that are misclassified by a weak learner?

A. Increased

B. Decreased

C. Randomly assigned

D. Remains unchanged

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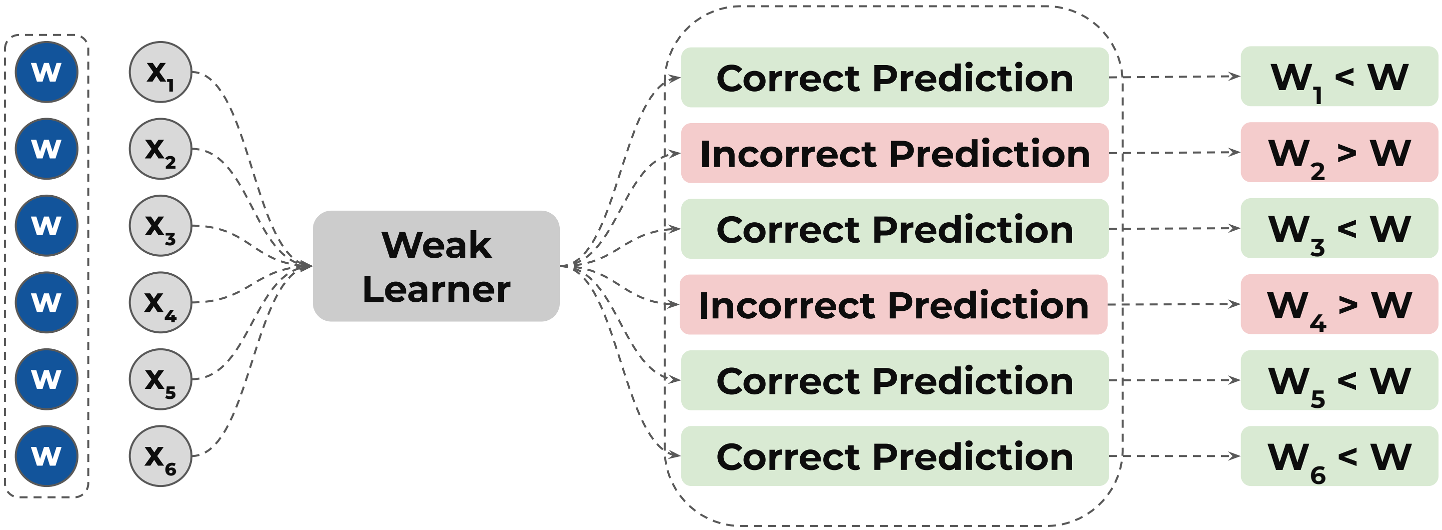
AdaBoost - Weight Updates

Weights are **increased** for the **misclassified instances** and **decreased** for the **correctly classified instances**



Boosting Quiz

What role does the learning rate play in Gradient Boosting Models?

A. Adjusts the depth of trees

B. Controls the contribution of each tree to the final ensemble

C. Defines the number of weak learners to be trained

D. Determines the fraction of data used

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Learning Rate in Gradient Boost

Learning rate (η) is a **small positive number** that **controls the contribution of each weak learner** to the ensemble

It controls the **magnitude of the steps** taken by the ensemble towards **an optimal solution**

0.0 **η** **∞**

$$\hat{Y}_i = \hat{Y}_{i-1} + \eta \cdot h_i$$

η is **generally** set to smaller values **(0.01 to 0.2)** to take small steps towards the optimal solution and avoid overfitting

The **learning rate** is **multiplied** with the **predicted residual (h_i)** from each weak learner **added to the final ensemble**, thereby **dictating their contribution** by **adjusting the magnitude** of the predicted residual

Boosting Quiz

What is the primary difference between the ways Gradient Boost and AdaBoost improve their performance with each iteration?

A. Gradient Boosting minimizes the residual errors, while AdaBoost updates sample weights

B. Gradient Boosting updates sample weights, while AdaBoost minimizes residual errors

C. Both methods update sample weights

D. Both methods minimize residual errors

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AdaBoost V/S Gradient Boost

AdaBoost	Gradient Boost
Begins with equal importance for all data points	Begins with an initial learner that makes simple predictions, like log(odds) value or a simple decision tree
After training, the algorithm increases importance of misclassified instances	After training, the algorithm focuses on minimizing residuals (difference between observed and predicted values)
Subsequent learners focus more on these challenging instances	Subsequent learners focus on learning patterns in the residuals from the previous learners
The final model combines weak learners by assigning higher importance to learners that perform well on challenging instances	The final model combines weak learners by adding them sequentially, with each new learner reducing the errors made by the previous ones

Boosting Quiz

In XGBoost, the parameter _____ controls the step size shrinkage used to prevent overfitting, _____ specifies the minimum loss reduction required to make a further partition on a leaf node, and _____ is used to handle class imbalance by controlling the balance between positive and negative classes.

A `learning_rate, gamma, scale_pos_weight`

B `gamma, learning_rate, scale_pos_weight`

C `gamma, learning_rate, scale_pos_weight`

D `scale_pos_weight, gamma, learning_rate`

Boosting Quiz

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C. `gamma, learning_rate, scale_pos_weight`

D. `scale_pos_weight, gamma, learning_rate`

Parameters of XGBoost

Parameters	Description
`learning_rate`	Gradient boosted decision trees are quick to learn and overfit training data. One effective way to slow down learning in the model is to use a learning rate, also called shrinkage
`gamma`	A node is split only when the resulting split gives a positive reduction in the loss function. Gamma specifies the minimum loss reduction required to make a split. Higher the gamma value, the lesser the chances of overfitting
`scale_pos_weight`	Control the balance of positive and negative weights. It can have any positive value as input. It helps in imbalanced classification problems
`colsample_bytree`	Proportion of features randomly sampled to train each individual tree in XGBoost
`colsample_bylevel`	Fraction of features randomly sampled at each tree level when building an XGBoost model

Great Learning

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